

Effective-temperature multipoles for relativistic transport.

V. Intensity semantics inside polarised radiative transfer

Jiwon Park,^{1,*} Myung-Ki Cheoun,^{1,†} and Jubin Park^{1,‡}

¹*Department of Physics and OMEG Institute, Soongsil University, Dongjak-gu, Seoul 06978, Republic of Korea*
(Dated: April 3, 2026)

Papers I–IV developed the effective-temperature (Teff) multipole representation for the scalar intensity sector of relativistic radiation. Polarised radiative transfer operates on the distribution matrix $\mathcal{P}_{ab}(\hat{e})$, whose trace is the scalar intensity I and whose traceless parts give the E - and B -mode polarisation. The present paper establishes the minimal bridge between the Teff intensity language and the standard polarisation hierarchy. We show that the scalar intensity trace inherits the Teff semantics of Papers I and IV without modification, and that the intensity quadrupole—which sources linear polarisation via Thomson scattering—is given exactly by $I_{ab} = c_\xi \langle \Theta^4 e_{\langle a} e_{\rangle b} \rangle_\Omega$, a statistics-independent nonlinear functional of the Teff multipoles (Proposition 1). When the Teff dipole T_a is nonzero, the quartic composition Θ^4 generates an $\ell = 2$ contribution proportional to A^2 that is absent from the linear approximation and can dominate the actual Thomson source at moderate dipole amplitudes: in the axisymmetric worked example $(A, Q) = (0.30, 0.15)$, the linear bridge underestimates by 52.5%. The polarisation multipoles E_{A_ℓ} and B_{A_ℓ} are not parameterised by Teff; they remain in the standard PSTF hierarchy. The paper defines the scope boundary: Teff provides intensity-side transparency and a nonlinear source parameterisation; the spin-2 polarisation sector retains its independent structure. No claim is made about a full Teff parameterisation of the spin-2 polarisation sector or about observed-sky E/B phenomenology; the result is an intensity-side scope theorem and source-term bridge. We also extend the bridge to the (Θ, η) two-field manifold of Papers II–III: the direction-dependent chemical potential $\eta(\hat{e})$ breaks the statistics independence of the one-field bridge, but its pure dipole component decouples from the Thomson source at linear order; the first η -induced correction enters at $O(A_\eta^2, A A_\eta)$; practical smallness of the η -correction requires $2 C_\xi A_\eta / A \ll 1$.

I. INTRODUCTION

The cosmic microwave background (CMB) carries a rich polarisation signal generated by Thomson scattering of anisotropic radiation off free electrons at recombination. The intensity quadrupole I_{ab} is the sole source of linear E -mode polarisation at leading order; its amplitude and angular structure determine the shape of the EE power spectrum on scales smaller than the Hubble radius at decoupling. In the standard Boltzmann hierarchy [1, 2], I_{ab} is a single PSTF multipole coefficient with no internal decomposition: its dependence on the lower multipoles I_0 (monopole) and I_a (dipole) is encoded implicitly in the coupled hierarchy.

Papers I–III [3–5] established the one-field and two-field Teff manifold, tangency diagnostic, and entropy geometry for the scalar intensity sector. Paper IV [6] developed the operational observable semantics. The present paper uses only the intensity-side structure required for the scalar trace. All four papers remain in the scalar (spin-0) sector; Papers II–III extend it by incorporating number-density and chemical-potential structure through the direction-dependent $\eta(\hat{e})$. The question motivating the present paper is: how does this intensity-side

formalism interface with the standard polarised Boltzmann hierarchy?

The answer has two parts. First, the scalar intensity trace $I = \text{tr}(\mathcal{P})$ inherits the Teff semantics without modification— $T_{\text{eff}}(\hat{e})$ is the per-direction brightness temperature for Planck statistics, and the tangency diagnostic monitors the intensity sector. Second, the intensity quadrupole that sources Thomson polarisation is given by the *exact nonlinear functional* $I_{ab} = c_\xi \langle \Theta^4 e_{\langle a} e_{\rangle b} \rangle_\Omega$ (Proposition 1). The nonlinear character is physically significant: when the Teff dipole T_a is nonzero, the quartic composition Θ^4 generates an $\ell = 2$ contribution proportional to A^2 that is absent from the linear approximation $I_{ab} \approx 4c_\xi T_{ab}$. At moderate dipole amplitude $(A, Q) = (0.30, 0.15)$, this dipole-squared term accounts for more than half the Thomson source, and the linear bridge underestimates by 52.5%.

We do not introduce a matrix-valued effective temperature, extend Teff to the full polarisation tensor, or couple the framework to a polarised transport solver. The contribution is a scope theorem and a source-term bridge.

Three contributions: **C1** (Intensity inheritance, §III A): the scalar intensity trace inherits the Teff semantics. **C2** (Nonlinear source bridge, §III B): Proposition 1 gives the exact Thomson source as a functional of the Teff multipoles. **C3** (Spin-2 boundary, §III D): the E_{A_ℓ} and B_{A_ℓ} hierarchies are not parameterised by Teff.

* cosmosapjw@soongsil.ac.kr

† cheoun@ssu.ac.kr

‡ holonov@ssu.ac.kr

II. STANDARD POLARISED RADIATIVE TRANSFER

A. Polarisation tensor and Stokes decomposition

Following Challinor [1, 2], the polarised radiation field on a spacetime $(\mathcal{M}, g_{ab})$ with timelike observer u^a is described by the polarisation tensor $\mathcal{P}_{ab}(x, E, \hat{e})$ defined on the screen space $S_{ab} := h_{ab} - e_a e_b$ perpendicular to both the propagation direction e^a and the observer four-velocity u^a (where $h_{ab} = g_{ab} + u_a u_b$). The tensor decomposes as

$$\mathcal{P}_{ab} = \frac{1}{2} I S_{ab} + \mathcal{P}_{ab}^E + \mathcal{P}_{ab}^B, \quad (1)$$

where $I = S^{ab} \mathcal{P}_{ab}$ is the scalar intensity (the trace), $\mathcal{P}_{ab}^E$ is the electric-parity (gradient) polarisation, and $\mathcal{P}_{ab}^B$ is the magnetic-parity (curl) polarisation. In the Stokes parameterisation, I is the total intensity, Q and U describe linear polarisation (screen-basis-dependent), and V describes circular polarisation (zero for Thomson scattering in the non-relativistic electron limit).

B. PSTF multipole hierarchies

The angular dependence of I , $\mathcal{P}^E$, and $\mathcal{P}^B$ is expanded in projected symmetric trace-free (PSTF) multipoles [1]. For the scalar intensity:

$$I(x, E, \hat{e}) = \sum_{\ell \geq 0} I_{A_\ell}(x, E) e^{A_\ell}, \quad (2)$$

where I_{A_ℓ} are rank- ℓ PSTF tensors. The polarisation sector follows the standard covariant polarised Boltzmann hierarchy notation of Challinor [1, 2]: the electric-parity multipoles E_{A_ℓ} ($\ell \geq 2$) and magnetic-parity multipoles B_{A_ℓ} ($\ell \geq 2$) are rank- ℓ PSTF tensors carrying additional spin-2 screen-space structure; they have no scalar Teff analog. We do not reproduce the full polarisation-hierarchy equations here, as the present paper uses only the scalar intensity expansion (2) and the Thomson coupling to the intensity quadrupole I_{ab} .

The normalisation convention follows Paper I and Challinor [1]: the ℓ -th intensity multipole is $I_{A_\ell} = \Delta_\ell^{-1} \int I(\hat{e}) e_{\langle A_\ell \rangle} d\Omega$ with $\Delta_\ell = 4\pi 2^\ell (\ell!)^2 / (2\ell+1)!$ (Paper I, Eq. (14); the lowest values are $\Delta_0 = 4\pi$, $\Delta_1 = 4\pi/3$, $\Delta_2 = 8\pi/15$). In the axisymmetric case, the scalar quadrupole coefficient reduces to the standard Legendre projection $I_2 = (2\ell+1)/2 \int I(\mu) P_\ell(\mu) d\mu$, which is independent of the 3D PSTF tensor normalization. Other conventions (Hu–White, Zaldarriaga–Seljak) differ by factors of $\sqrt{6}$ or $4\pi/(2\ell+1)$; the structural results below are convention-independent.

C. Thomson scattering source

At linear order in the electron rest frame, Thomson scattering couples the intensity to polarisation through

the intensity quadrupole [1]:

$$\dot{E}_{ab}|_{\text{Thomson}} = -\frac{1}{10} n_e \sigma_T I_{ab}, \quad (3)$$

where n_e is the electron number density and σ_T the Thomson cross-section. This is the *only* intensity-to-polarisation coupling at leading order: the intensity monopole and dipole do not source E -modes; higher intensity multipoles ($\ell \geq 3$) contribute only at higher perturbative orders. B -mode multipoles are sourced by tensor metric perturbations or gravitational lensing, not by intensity.

The coefficient $-1/10$ follows from the Challinor normalisation: the Thomson source integrated over the angular distribution of incoming radiation yields $-(3n_e \sigma_T / 16\pi) \int I(\hat{e}') e'_{\langle a \rangle} e'_{\langle b \rangle} d\Omega'$, which in the PSTF normalisation gives $-(1/10)n_e \sigma_T I_{ab}$. This structural result—the intensity quadrupole as the sole linear source of E -mode polarisation—is the foundation for the Teff bridge.

III. TEFF INSIDE POLARISED RADIATIVE TRANSFER

A. Intensity sector: inheritance from Papers I and IV

Within the decomposition (1), the scalar intensity trace I satisfies the same Boltzmann hierarchy as the scalar distribution f of Papers I–IV. The Teff ansatz $f = \Phi_\xi(y/\Theta(\hat{e}))$ parameterises I along each direction: the energy-integrated intensity is

$$I(\hat{e}) = c_\xi T_0^4 \Theta(\hat{e})^4, \quad (4)$$

where $c_\xi = \int_0^\infty z^3 \Phi_\xi(z) dz$ is the statistics-dependent Stefan–Boltzmann prefactor. The scalar intensity trace inherits the Teff semantics of the one-field intensity sector: T_0 is the occupation temperature, $T_{\text{eff}}(\hat{e}) = T_0 \Theta(\hat{e})$ is the per-direction temperature, and for blackbody photons (Planck distribution with $\mu = 0$) this is the exact brightness temperature at all frequencies. For quantum statistics more generally, $T_{\text{eff}}(\hat{e})$ is the occupation temperature. The tangency diagnostic $\mathcal{D}_{\geq 2}$ monitors the intensity sector and provides the error bound of Paper IV, Theorem 3.

What Teff does *not* do inside the polarised hierarchy: it does not parameterise the traceless parts $\mathcal{P}_{ab}^E$ or $\mathcal{P}_{ab}^B$. These carry independent spin-2 information with their own evolution equations; their dynamical content is not reducible to the scalar $\Theta(\hat{e})$.

B. Teff quadrupole as Thomson source: exact nonlinear bridge

The intensity quadrupole in the Teff representation is obtained by integrating (4) against the PSTF kernel:

$$I_{ab} = c_\xi T_0^4 \langle \Theta(\hat{e})^4 e_{\langle a} e_{\rangle b} \rangle_\Omega. \quad (5)$$

Proposition 1 (Exact nonlinear Teff–Thomson bridge). *Let $\Theta(\hat{e}) > 0$ with $\langle \Theta \rangle_\Omega = 1$, and let I_{ab} denote the $\ell = 2$ PSTF intensity multipole with energy weight y^3 . Then:*

- (i) *The exact bridge is $I_{ab} = c_\xi T_0^4 \langle \Theta^4 e_{\langle a} e_{\rangle b} \rangle_\Omega$.*
- (ii) *The Thomson polarisation source is*

$$\dot{E}_{ab}|_{\text{Thomson}} = -\frac{c_\xi T_0^4}{10} n_e \sigma_T \langle \Theta^4 e_{\langle a} e_{\rangle b} \rangle_\Omega. \quad (6)$$

(iii) *Both expressions are computable by angular quadrature from the Teff multipoles $\{T_{A\ell}\}$ without additional modelling.*

Proof. Part (i) follows directly from the definition (5) and the Teff ansatz (4). Part (ii) substitutes (i) into the Thomson source (3). Part (iii) notes that $\Theta(\hat{e})$ is determined by the finite set of Teff multipoles $\{T_{A\ell}\}$, and the angular integral is a standard quadrature over the momentum sphere. $\square$

Corollary 2 (Statistics independence). *The ratio $I_{ab}/(c_\xi T_0^4) = \langle \Theta^4 e_{\langle a} e_{\rangle b} \rangle_\Omega$ is a pure angular-geometric quantity, independent of the statistics parameter ξ . The prefactor c_ξ enters only as an overall multiplicative scale.*

C. Weak-anisotropy expansion and the dipole-squared term

Writing $\Theta = 1 + \theta$ with $\langle \theta \rangle_\Omega = 0$ (gauge condition), the quartic $\Theta^4 = (1 + \theta)^4$ expands as

$$\Theta^4 = 1 + 4\theta + 6\theta^2 + 4\theta^3 + \theta^4. \quad (7)$$

Projecting each term onto $\ell = 2$ (the PSTF trace-free quadrupole):

Zeroth order. $\langle 1 \cdot e_{\langle a} e_{\rangle b} \rangle_\Omega = 0$ by the trace-free property.

First order (the linear bridge). $4\langle \theta e_{\langle a} e_{\rangle b} \rangle_\Omega = 4T_{ab}$, where T_{ab} is the $\ell = 2$ PSTF multipole of Θ . This gives the standard linear relation $I_{ab} \approx 4c_\xi T_0^4 T_{ab}$.

Second order (the dipole-squared correction). $6\langle \theta^2 e_{\langle a} e_{\rangle b} \rangle_\Omega$. For the axisymmetric $L = 2$ truncation $\theta = A P_1(\mu) + Q P_2(\mu)$:

$$\theta^2 = A^2 \mu^2 + 2AQ \mu P_2(\mu) + Q^2 P_2(\mu)^2. \quad (8)$$

The key term is $A^2 \mu^2 = A^2(\frac{1}{3} + \frac{2}{3} P_2(\mu))$: the dipole squared projects onto $\ell = 0$ (absorbed into the monopole) and $\ell = 2$ (contributing to I_{ab}). The $\ell = 2$ projection of the full θ^2 is

$$\langle \theta^2 e_{\langle a} e_{\rangle b} \rangle_\Omega = \left(\frac{2A^2}{3} + \frac{2AQ}{7} + \frac{2Q^2}{7} \right) \hat{e}_{2,ab}, \quad (9)$$

where $\hat{e}_{2,ab}$ is the unit $\ell = 2$ PSTF tensor. The full weak-anisotropy expansion of the intensity quadrupole is therefore

$$\frac{I_{ab}}{c_\xi T_0^4} = 4T_{ab} + 6\langle \theta^2 e_{\langle a} e_{\rangle b} \rangle_\Omega + \mathcal{O}(\theta^3). \quad (10)$$

Physical significance. At $(A, Q) = (0.30, 0.15)$: the linear term contributes $4Q \times (2/5) = 0.24$ to the normalised $I_2/c_\xi T_0^4$, while the dipole-squared term contributes $6 \times (2A^2/3) \times (2/5) = 6 \times 0.06 \times 0.4 = 0.144$. The quadratic correction is 60% of the linear term. The full nonlinear value (by quadrature) is $I_2/c_\xi T_0^4 = 0.842$, while the linear prediction gives 0.400—a 52.5% underestimate (Table II).

This large discrepancy arises because the dipole-squared term $\propto A^2$ adds constructively to the intrinsic quadrupole term $\propto Q$. At $A = 0.30$, the dipole-squared contribution exceeds the intrinsic-quadrupole contribution: the Thomson source is dominated by the nonlinear mixing of the temperature dipole, not by the temperature quadrupole itself.

D. The spin-2 boundary

Three structural boundaries delimit the reach of Teff in the polarised hierarchy.

First: E-mode sourcing. The E -mode multipoles $E_{A\ell}$ are sourced by I_{ab} —which Teff parameterises through Proposition 1—but evolve via their own coupled hierarchy. Teff does not predict $E_{A\ell}$; it provides a transparent parameterisation of the source that drives their evolution. The diagnostic $\mathcal{D}_{\geq 2}$ monitors the intensity-sector contribution to the source as a conditional tangency proxy (Paper IV, Theorem 3, under assumptions H1–H3), but says nothing about the accuracy of the subsequent E -mode propagation.

Second: B-mode independence. The B -mode multipoles $B_{A\ell}$ require tensor metric perturbations (gravitational waves) or gravitational lensing for generation. Neither mechanism involves the intensity quadrupole at leading order. Teff has no bearing on B -mode physics.

Third: screen-basis information. The Stokes parameters Q and U depend on the choice of polarisation basis on the screen space perpendicular to $\hat{e}$ and u^a . The scalar $\Theta(\hat{e})$ carries no screen-basis information: it is invariant under screen rotations, while $Q \pm iU$ transforms as a spin-2 field. The passage from the covariant $E_{A\ell}$ to the screen-dependent $Q \pm iU$ involves a spin-2 projection that lives entirely outside the Teff framework.

Remark 3 (Scope statement). Teff parameterises one object—the scalar intensity trace $I = \text{tr}(\mathcal{P})$ —and through Proposition 1 provides a transparent, nonlinear parameterisation of the Thomson source I_{ab} . The $E_{A\ell}$ and $B_{A\ell}$ hierarchies, the Stokes parameters, and the spin-2 geometry of the polarisation field are entirely outside its reach. A matrix-valued Teff parameterising the full $\mathcal{P}_{ab}$ would require a fundamentally different ansatz.

IV. CANONICAL THOMSON SOURCE EXAMPLE

We illustrate the bridge with the canonical polarisation-generating configuration: isotropic electrons scattering an anisotropic radiation field parameterised by the Teff multipoles (A, Q) .

Table II shows the intensity quadrupole $I_2 = (5/2)\langle\Theta^4 P_2\rangle$ as a function of (A, Q) , comparing the exact nonlinear value with the linear $(4Q)$ and quadratic $(4Q + 6\langle\theta^2 P_2\rangle)$ approximations.

Three regimes are visible.

Pure quadrupole ($A = 0$). The linear bridge works well: $< 5\%$ error at $Q \leq 0.15$. The dipole-squared term vanishes identically, and the quadratic correction comes only from Q^2 terms. At $Q = 0.30$, the linear error reaches 9.4% and the quadratic correction captures most of it (residual 1.6%).

Moderate dipole ($A = 0.15, Q = 0.10$). The dipole-squared contribution $\propto A^2$ enters at the same order as the intrinsic quadrupole: the linear bridge underestimates by 28.8%. The quadratic correction reduces the error to 2.8%. This is the regime typical of the semi-transparent zone in core-collapse supernovae and binary neutron star mergers.

Strong dipole ($A \geq 0.30$). The dipole-squared term dominates: at $(A, Q) = (0.30, 0.15)$, the linear bridge gives $I_2/c_\xi T_0^4 = 0.400$ while the exact value is 0.842—a 52.5% underestimate. At $(A, Q) = (0.40, 0.20)$: 60.9% underestimate. The quadratic correction captures most of the deficit (7.7% and 11.7% residual errors respectively), but the full nonlinear bridge (Proposition 1) is needed for percent-level accuracy.

Practical consequence. An analysis using the linear relation $I_{ab} \approx 4c_\xi T_{ab}$ and ignoring the dipole-squared contribution would underestimate the Thomson polarisation source by up to a factor of two at moderate beam-like configurations. The Teff formalism makes this nonlinear mixing explicit and computable from the same multipoles used for the intensity transport. The statistics independence (Corollary 2) ensures that the fractional error of the linear bridge is the same for photons (Planck), neutrinos (FD), and classical particles (MB).

V. TWO-FIELD EXTENSION OF THE THOMSON BRIDGE

Papers II–III introduced the two-field ansatz

$$f(x, \hat{e}) = \Phi_\xi \left(\frac{x}{\Theta(\hat{e})} - \eta(\hat{e}) \right), \quad x > 0, \quad (11)$$

in which a direction-dependent chemical potential $\eta(\hat{e})$ parameterises number–energy decoupling. Paper II (Theorem 12) showed that at scalar $\eta = \eta_0$ the spectral integral factorises: the energy-integrated intensity becomes $I(\hat{e}) = \mathcal{I}_\xi^{(3)}(\eta_0) T_0^4 \Theta(\hat{e})^4$, where $\mathcal{I}_\xi^{(p)}(\eta) =$

$\int_0^\infty u^p \Phi_\xi(u - \eta) du$. At scalar η , the factor $\mathcal{I}_\xi^{(3)}(\eta_0)$ replaces c_ξ in Proposition 1 and the statistics independence of Corollary 2 is preserved. When η varies with direction, this factorisation fails and the intensity quadrupole acquires a new structure.

A. Exact two-field intensity bridge

Proposition 4 (Exact two-field intensity bridge). *Under the two-field ansatz (11) with $\Theta(\hat{e}) > 0$ and, for BE, $\eta(\hat{e}) \leq 0$ (Paper II, Theorem 1), the energy-integrated intensity trace is*

$$I(\hat{e}) = T_0^4 \Theta(\hat{e})^4 \mathcal{I}_\xi^{(3)}(\eta(\hat{e})), \quad (12)$$

and the intensity quadrupole is

$$I_{ab}^{(2f)} = T_0^4 \langle \Theta^4 \mathcal{I}_\xi^{(3)}(\eta) e_{\langle a} e_{b \rangle} \rangle_\Omega. \quad (13)$$

When η varies with direction, the ratio $I_{ab}^{(2f)}/\mathcal{I}_\xi^{(3)}(\eta_0)$ is no longer a pure angular-geometric quantity: the statistics independence of Corollary 2 is broken.

Proof. The substitution $u = x/\Theta$ gives $\int_0^\infty x^3 \Phi_\xi(x/\Theta - \eta) dx = \Theta^4 \int_0^\infty u^3 \Phi_\xi(u - \eta) du = \Theta^4 \mathcal{I}_\xi^{(3)}(\eta)$. Since $\eta = \eta(\hat{e})$, the factor $\mathcal{I}_\xi^{(3)}(\eta)$ remains inside the angular average when projecting onto $e_{\langle a} e_{b \rangle}$. $\square$

At scalar $\eta = \eta_0$, $\mathcal{I}_\xi^{(3)}$ exits the angular integral and (13) reduces to Proposition 1 with $c_\xi \rightarrow \mathcal{I}_\xi^{(3)}(\eta_0)$. For Maxwell–Boltzmann statistics, $\mathcal{I}_\xi^{(3)}(\eta) = 6 e^\eta$, so the $e^{\eta(\hat{e})}$ factor enters multiplicatively; ratios of same-sector observables remain η -independent (Paper II, Remark 15). For Fermi–Dirac and Bose–Einstein, the functional form of $\mathcal{I}_\xi^{(3)}$ is nonlinear in η and no such cancellation occurs in general.

B. Pure η -dipole linear decoupling

Proposition 5 (Linear η -dipole Thomson decoupling). *In the axisymmetric ($L_\Theta = 2, L_\eta = 1$) parameterisation*

$$\Theta(\mu) = 1 + A P_1(\mu) + Q P_2(\mu), \quad \eta(\mu) = \eta_0 + A_\eta P_1(\mu), \quad (14)$$

the pure η -dipole does not contribute to the intensity quadrupole at linear order: expanding (12) about the isotropic background at $O(\theta, \delta\eta)$ yields

$$\langle \delta\eta(\hat{e}) e_{\langle a} e_{b \rangle} \rangle_\Omega = 0 \quad (15)$$

because $\delta\eta = A_\eta P_1(\mu)$ is an $\ell = 1$ harmonic and the quadrupole projector selects $\ell = 2$ only.

Proof. To linear order, $\Theta^4 \mathcal{I}_\xi^{(3)}(\eta) \approx \mathcal{I}_\xi^{(3)}(\eta_0)[1 + 4\theta] + 3\mathcal{I}_\xi^{(2)}(\eta_0)\delta\eta + O(2)$, where $\theta = \Theta - 1$ and $\delta\eta = \eta - \eta_0$, using $d\mathcal{I}_\xi^{(3)}/d\eta = 3\mathcal{I}_\xi^{(2)}$ (Paper II). The η -dependent linear term is $3\mathcal{I}_\xi^{(2)}(\eta_0)A_\eta P_1(\mu)$. Since $\langle P_1 e_{(a} e_{b)} \rangle_\Omega = 0$ by Clebsch–Gordan selection ($P_1 \otimes P_2 \rightarrow \ell = 1, 3$), this term vanishes upon $\ell = 2$ projection. $\square$

This result is specific to ($L_\eta = 1$): if η carried a quadrupole component $Q_\eta P_2(\mu)$, the linear term $3\mathcal{I}_\xi^{(2)}(\eta_0)Q_\eta P_2$ would enter I_{ab} directly. The physical content matches Paper II’s identification of the η -dipole as a *number*-sector observable whose leading signature is in f_N/f_E (Proposition 17 therein), not in the intensity quadrupole.

C. Quadratic onset of η -induced correction

Proposition 6 (Quadratic η –Thomson correction). *Under the same axisymmetric parameterisation, the first nonzero η -induced contribution to the intensity quadrupole appears at quadratic order. In the dipole-only sector ($Q = 0$), the quadrupole scalar coefficient receives three terms:*

$$\left. \frac{I_2}{T_0^4} \right|_{Q=0}^{O(2)} = 4\mathcal{I}_\xi^{(3)}A^2 + 8\mathcal{I}_\xi^{(2)}AA_\eta + 2\mathcal{I}_\xi^{(1)}A_\eta^2, \quad (16)$$

where all spectral integrals are evaluated at η_0 . The first term is the one-field dipole-squared contribution of Proposition 1; the second and third are the mixed and pure η -dipole corrections, respectively.

Proof. Expanding $\Theta^4 \mathcal{I}_\xi^{(3)}(\eta)$ to second order:

$$\begin{aligned} \Theta^4 \mathcal{I}_\xi^{(3)}(\eta) &\approx \mathcal{I}_\xi^{(3)} + 4\mathcal{I}_\xi^{(3)}\theta + 6\mathcal{I}_\xi^{(3)}\theta^2 \\ &\quad + 3\mathcal{I}_\xi^{(2)}\delta\eta + 12\mathcal{I}_\xi^{(2)}\theta\delta\eta + 3\mathcal{I}_\xi^{(1)}(\delta\eta)^2, \end{aligned} \quad (17)$$

using $d^2\mathcal{I}_\xi^{(3)}/d\eta^2 = 3 \cdot 2\mathcal{I}_\xi^{(1)}$. Setting $\theta = AP_1$, $\delta\eta = A_\eta P_1$, the $\ell = 2$ content arises from $P_1^2 = \frac{1}{3} + \frac{2}{3}P_2$: the θ^2 term gives $4\mathcal{I}_\xi^{(3)}A^2 P_2$, the $\theta\delta\eta$ term gives $8\mathcal{I}_\xi^{(2)}AA_\eta P_2$, and the $(\delta\eta)^2$ term gives $2\mathcal{I}_\xi^{(1)}A_\eta^2 P_2$. $\square$

The ratio of the mixed correction to the one-field dipole-squared term is

$$\frac{8\mathcal{I}_\xi^{(2)}AA_\eta}{4\mathcal{I}_\xi^{(3)}A^2} = 2C_\xi(\eta_0)\frac{A_\eta}{A}, \quad (18)$$

where $C_\xi = \mathcal{I}_\xi^{(2)}/\mathcal{I}_\xi^{(3)}$ is the cross-sector ratio of Paper II. For moderate Fermi–Dirac degeneracy ($\eta_0 \sim 3$), $C_{\text{FD}} \approx 0.25$; with $A_\eta \sim A$, the mixed term reaches $\sim 50\%$ of the one-field term. The pure η -dipole term is suppressed by a further factor $\mathcal{I}_\xi^{(1)}/\mathcal{I}_\xi^{(2)} \approx 0.32$ at $\eta_0 = 3$.

Corollary 7 (One-field bridge: quadratic η -correction). *On the minimal two-field manifold with ($L_\eta = 1$) η -dipole, the first η -induced correction to the intensity quadrupole enters at quadratic order $O(A_\eta^2, AA_\eta)$ and vanishes at linear order (Proposition 5). The ratio of the mixed correction to the one-field dipole-squared term is $2C_\xi(\eta_0)A_\eta/A$ (Eq. (18)), which is ~ 0.5 at moderate FD degeneracy with $A_\eta \sim A$. Practical smallness of the η -correction therefore requires $2C_\xi A_\eta/A \ll 1$, not merely $A_\eta < A$.*

D. Conditional polarised extension

If the polarisation tensor factorises as

$$\mathcal{P}_{ab}(x, \hat{e}) = \frac{1}{2}\Phi_\xi\left(\frac{x}{\Theta} - \eta\right)[S_{ab}(\hat{e}) + \varepsilon_{ab}(\hat{e})], \quad (19)$$

with $S^{ab}\varepsilon_{ab} = 0$ and $|\varepsilon| < 1$, then the trace $I = S^{ab}\mathcal{P}_{ab}$ reduces exactly to (12): the scalar intensity inherits the two-field bridge of Proposition 4, and the Thomson source $\dot{E}_{ab}|_{\text{Thomson}} = -(n_e\sigma_T/10)I_{ab}^{(2f)}$ follows.

Around an unpolarised background ($\varepsilon_{ab} = 0$), the scalar variations ($\delta\Theta, \delta\eta$) and the spin-2 variation $\delta\varepsilon_{ab}$ do not cross-contaminate the scalar trace at linear order, because $S^{ab}\delta\varepsilon_{ab} = 0$ prevents cross-contamination of the trace. Beyond linear order, or at finite $|\varepsilon|$, the coupling between trace and traceless sectors requires the full tensor inner product and is not addressed here.

E. Open structural questions

Three questions remain open for the two-field polarised manifold. (i) Does the factorised ansatz (19) define a finite-dimensional exponential family in a matrix-valued information geometry, and if so, does Paper III’s Gram–Hessian identity survive in block-diagonal form? (ii) Does a Fisher-orthogonal Pythagorean decomposition extend from the scalar trace sector to the full $(\Theta, \eta, \varepsilon_{ab})$ state space? (iii) Can one construct a symmetrising thermodynamic potential for the coupled intensity–polarisation hierarchy, or does the spin-2 sector introduce fundamentally new obstructions? These questions require a new ansatz class and a tensor-valued collision geometry beyond the scope of the present series.

VI. DISCUSSION

A. What Teff buys

Teff is not a polarisation formalism; it is an intensity-side source formalism. Within the polarised Boltzmann hierarchy, Teff parameterises one object—the scalar intensity trace I —and through the nonlinear bridge

(Proposition 1) provides a transparent, temperature-based parameterisation of the Thomson polarisation source I_{ab} .

Two advantages in the polarised context. First, *semantic*: the Thomson source is expressed in terms of the direction-dependent temperature field whose operational meaning was established in Paper IV. An observer reading $T_{\text{eff}}(\hat{e}) = T_0\Theta(\hat{e})$ from the Teff reconstruction obtains a direct physical observable (brightness temperature for Planck statistics); the Thomson source follows as a nonlinear functional of this same field. Second, *structural*: the nonlinear bridge (10) decomposes the intensity quadrupole into contributions from each Teff multipole, making the dipole-squared $\rightarrow$ quadrupole coupling explicit. In the standard PSTF hierarchy, I_{ab} is a single coefficient with no such decomposition.

B. What Teff does not provide

The $E_{A\ell}$ and $B_{A\ell}$ hierarchies are untouched. Stokes Q and U carry spin-2 screen-basis information that no scalar $\Theta(\hat{e})$ can encode. Teff is not a replacement for the standard polarised Boltzmann hierarchy; it is a natural-parameter language for the intensity side of that hierarchy, with a specific and computable bridge to the Thomson source.

The diagnostic $\mathcal{D}_{\geq 2}$ of Paper I monitors the *intensity* sector; it does not certify the accuracy of the polarisation evolution. A small $\mathcal{D}_{\geq 2}$ guarantees that the intensity-side contribution to the Thomson source is well-captured, but the subsequent propagation of E -modes through the coupled hierarchy introduces additional errors that lie outside the diagnostic's scope.

C. Deferred extensions

Three extensions are deferred. First, a matrix-valued effective temperature parameterising the full $\mathcal{P}_{ab}$ would require a fundamentally different ansatz—the traceless

spin-2 part has no scalar analog—and is outside the current scope. Second, the intensity-side two-field bridge (§V) resolves part of the (Θ, η) connection; the remaining open question is whether the full entropy geometry (Paper III's Gram–Hessian identity, Pythagorean decomposition) survives on the polarised $(\Theta, \eta, \varepsilon_{ab})$ manifold (§V, open questions). Third, all results here are instantaneous algebraic relations valid at a single spacetime point; coupling the Teff-parameterised intensity to a polarised transport solver and evaluating the diagnostic on dynamically evolved states is the subject of future work.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea

TABLE I. Statistics independence of I_{ab}/c_ξ at $(A, Q) = (0.20, 0.15)$. Machine-precision agreement ($< 10^{-10}$).

Statistics	c_ξ	I_2/c_ξ
MB	6.000	0.6058545455
FD	5.682	0.6058545455
BE	6.494	0.6058545455

TABLE II. Intensity quadrupole: exact vs perturbative. $I_2/c_\xi = (5/2) \int_{-1}^1 \Theta^4 P_2 d\mu$ (axisymmetric Legendre projection). Lin. error: $|1 - I_2^{\text{lin}}/I_2^{\text{exact}}|$. At $(A, Q) = (0.30, 0.15)$, the linear bridge underestimates by 52.5%.

A	Q	Exact	Linear	Lin. err	Quad. err
0	0.05	0.135	0.133	1.5%	0.05%
0	0.15	0.419	0.400	4.5%	0.4%
0.15	0.10	0.375	0.267	28.8%	2.8%
0.30	0.15	0.842	0.400	52.5%	7.7%
0.40	0.20	1.363	0.533	60.9%	11.7%

government (MSIT).

-
- [1] A. Challinor, Microwave background polarization in cosmological models, Phys. Rev. D **62**, 043004 (2000).
 - [2] A. Challinor and A. Lasenby, CMB anisotropies in the CDM model: a covariant and gauge-invariant approach, Astrophys. J. **513**, 1 (1999).
 - [3] J. Park, M.-K. Cheoun, and J. Park, Effective-temperature multipoles. I, Phys. Rev. E (2026), paper I.
 - [4] J. Park, M.-K. Cheoun, and J. Park, Effective-temperature multipoles. II, Phys. Rev. E (2026), paper II.
 - [5] J. Park, M.-K. Cheoun, and J. Park, Effective-temperature multipoles. III, Phys. Rev. E (2026), paper III.
 - [6] J. Park, M.-K. Cheoun, and J. Park, Effective-temperature multipoles. IV, Phys. Rev. D (2026), paper IV.

TABLE III. Source translation: standard vs Teff formalism.

Aspect	Standard	Teff
Intensity mono.	I_0	$c_\xi T_0^4 \langle \Theta^4 \rangle_\Omega$
Intensity quad.	I_{ab}	$c_\xi \langle \Theta^4 e_{\langle a} e_{\langle b} \rangle} \rangle_\Omega$
Thomson source	$-\frac{1}{10} n_e \sigma_T I_{ab}$	Explicit Teff functional
$A^2 \rightarrow$ quad.	Hidden	Explicit
E/B modes	Standard	Unchanged
Stokes Q, U	Screen-basis	Not parameterised
Diagnostic	None	$\mathcal{D}_{\geq 2}$ on intensity (conditional)